

MISRATA, Libyan Arab JamahiriyaWMO: **620160**Lat: **32.325N**Lon: **15.059E**Elev: **33**StdP: **100.93**Time Zone: **2.00 (E02)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.2	9.2	-2.3	3.1	17.7	-0.2	3.7	16.5	15.6	14.4	14.9	13.8	3.4	240	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	37.1	21.8	34.8	21.6	32.8	21.8	26.5	29.6	25.9	29.1	25.4	28.8	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.6	21.0	28.5	25.0	20.2	28.1	24.4	19.4	27.8	82.7	29.7	80.2	29.2	78.0	28.9	31.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	10.8	9.5	6.1	42.5	2.1	2.4	4.6	44.3	3.4	45.7	2.2	47.1	0.7	48.8
			3.2	28.9	1.3	1.7	2.3	30.1	1.5	31.1	0.8	32.1	-0.1	33.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.2	14.9	14.6	16.5	18.9	22.1	25.1	27.2	28.1	27.1	23.8	19.3	15.7	(6)
(7)		DBStd	5.47	1.76	2.52	2.90	2.99	3.24	2.91	2.08	1.58	1.89	2.54	2.47	2.38	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	421	108	109	74	24	2	0	0	0	0	0	16	88	(9)
(10)		CDD10.0	4070	152	130	201	268	376	454	533	561	513	426	279	177	(10)
(11)		CDD18.3	1451	2	6	17	42	120	204	274	303	263	168	45	7	(11)
(12)		CDH23.3	12915	7	35	129	263	768	1629	2821	3435	2571	1063	168	27	(12)
(13)	CDH26.7	4592	0	9	42	98	316	650	1007	1314	832	288	33	3	(13)	
(14)	Wind	WSAvg	4.6	4.7	5.1	5.1	5.2	5.0	4.4	4.1	3.9	4.3	4.2	4.2	4.7	(14)
(15)	Precipitation	PrecAvg	270	57	25	21	9	4	1	0	1	10	41	40	55	(15)
(16)		PrecMax	430	216	70	73	54	37	19	1	17	67	136	201	164	(16)
(17)		PrecMin	77	5	1	0	0	0	0	0	0	0	0	2	2	(17)
(18)		PrecStd	84	47	19	20	11	8	3	0	4	16	37	39	40	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.4	27.4	31.4	35.0	38.0	41.1	39.2	38.3	38.7	35.3	30.2	26.5	(19)
(20)			MCWB	13.2	14.5	15.6	16.6	20.4	21.7	22.1	22.8	21.8	19.6	16.9	14.1	(20)
(21)		2%	DB	21.2	23.2	26.6	29.5	33.2	35.6	35.8	35.0	34.5	32.2	27.1	22.9	(21)
(22)			MCWB	12.9	13.4	14.8	17.4	19.0	20.9	22.2	23.3	22.4	19.7	17.4	14.1	(22)
(23)		5%	DB	19.5	20.6	23.6	26.4	29.8	32.6	33.0	33.0	32.1	29.6	25.1	20.9	(23)
(24)			MCWB	13.3	13.3	14.7	16.1	19.0	20.7	22.4	23.5	23.1	20.6	17.4	14.4	(24)
(25)		10%	DB	18.2	18.5	21.0	23.6	27.1	30.3	30.8	31.3	30.4	27.7	23.6	19.6	(25)
(26)			MCWB	13.4	13.1	14.3	15.9	18.3	20.7	23.0	24.1	23.7	21.2	17.7	14.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	16.2	17.9	23.6	23.0	25.4	26.8	27.8	27.2	24.8	21.6	17.8	(27)
(28)			MCDWB	18.7	21.0	23.2	30.2	32.2	29.0	29.6	31.6	29.7	28.4	24.4	21.0	(28)
(29)		2%	WB	15.6	15.4	16.9	19.0	21.2	24.2	26.0	26.6	26.2	23.6	20.2	16.7	(29)
(30)			MCDWB	18.2	19.3	20.9	23.8	27.0	27.9	29.0	29.7	29.2	26.9	23.8	19.8	(30)
(31)		5%	WB	14.8	14.7	16.2	17.9	20.5	23.5	25.4	26.0	25.5	22.8	19.4	16.0	(31)
(32)			MCDWB	17.8	18.3	20.2	22.2	25.5	27.3	28.7	29.3	28.7	26.4	23.2	19.1	(32)
(33)		10%	WB	14.1	14.1	15.5	17.1	19.8	22.8	24.9	25.6	24.9	22.1	18.6	15.2	(33)
(34)			MCDWB	17.0	17.7	19.4	21.3	24.2	26.7	28.3	29.0	28.3	26.1	22.4	18.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.2	7.3	7.9	7.5	7.8	8.0	6.9	6.9	7.4	7.6	7.2	(35)	
(36)			MCDBR	9.7	11.6	13.2	13.9	13.5	13.3	12.0	10.7	10.9	10.9	10.0	9.5	(36)
(37)			MCWBR	5.4	5.3	5.3	5.2	4.1	3.9	4.3	3.8	4.1	4.2	5.2	5.7	(37)
(38)		5% WB	MCDWB	6.8	8.2	9.0	8.4	9.3	8.0	7.1	6.5	6.8	7.5	7.8	7.1	(38)
(39)	MCWBR		4.9	5.0	4.9	4.3	3.8	3.7	4.1	3.5	4.0	3.9	5.1	5.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.386	0.413	0.485	0.540	0.542	0.481	0.479	0.479	0.516	0.472	0.427	0.394	(40)	
(41)		taud	2.231	2.139	1.937	1.832	1.854	2.042	2.066	2.082	1.980	2.065	2.137	2.211	(41)	
(42)		Ebn at Noon	813	829	793	767	769	815	814	808	756	757	757	778	(42)	
(43)		Edh at Noon	114	137	181	209	207	171	166	161	171	146	124	111	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.98	3.80	4.97	6.06	6.77	7.34	7.36	6.64	5.45	4.16	3.19	2.61	(44)	
(45)		RadStd	0.16	0.21	0.25	0.23	0.29	0.26	0.11	0.17	0.18	0.22	0.12	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page